University of
South-Eastern Norway

EXAMINATION INFORMATION PAGE
Written examination

Subject code: **IIA4117**	Subject name: Model Predictive Control	
Examination date: 05.12.2022	Examination time from/to: 09:00 – 12:00	Total hours: 3
Responsible subject teacher: Roshan Sharma		
Campus: Porsgrunn	Faculty: Faculty of Technology	
No. of assignments: 3	No. of attachments: 0	No. of pages incl. front page and attachments: 3
Permitted aids: None, only pen or pencil		
Information regarding attachments: No attachments		
Comments: Please use ball pen. This makes it easier for the examiner to read your answers. This exam counts 60% for the final grade.		

Select the type of examination paper	Spreadsheets ☐	Line sheets ☒

CANDIDATES MUST THEMSELVES CHECK THAT ALL ASSIGNMENTS AND ATTACHMENTS ARE IN ORDER.

This exam counts 60% for the final grade.

Problem 1

Assume that you are newly hired as an automation engineer at a firm. You have been assigned the task of designing an MPC for controlling a process whose nonlinear model is given by,

$$\begin{aligned} \frac{dx}{dt} &= f(x, u, t) \\ y &= g(x, u, t) \end{aligned} \tag{1}$$

Here x, y and u represents the process states, outputs and the control inputs respectively. As a first try you designed a linear MPC based on a linearized model of the plant given by,

$$x_{k+1} = Ax_k + Bu_k \tag{2}$$

$$y_k = Cx_k \tag{3}$$

When you applied the linear MPC to the nonlinear model as the target or real system, you got the following simulation results as shown in Figure 1.

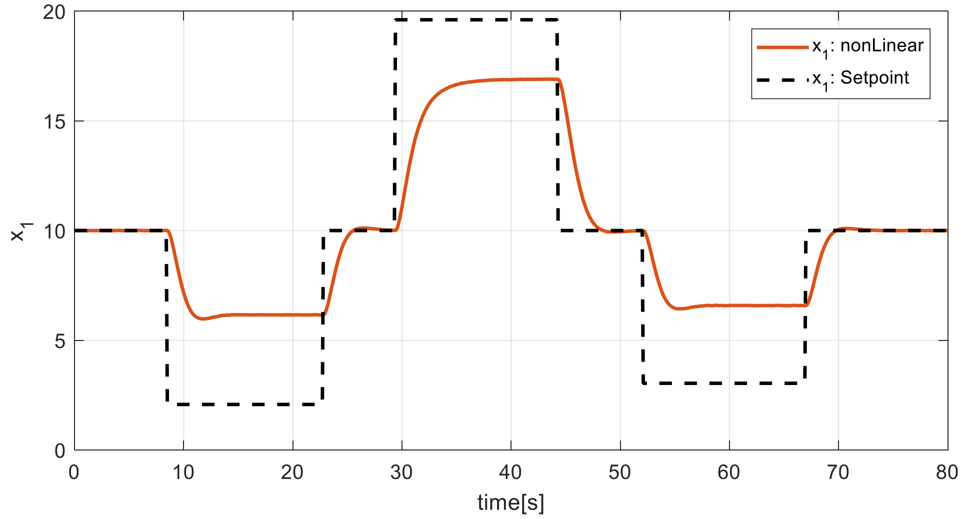

Figure 1: Output of the linear MPC when applied to the nonlinear plant model

You and your manager are not satisfied with the simulation results and want to improve the controller's performance. To do so, you plan to introduce integral action to your linear MPC by making use of the Δu formulation for integral action.

Tasks:

i) [8%] With the Δu formulation for obtaining integral action, an augmented model first has to be derived. Show clearly all the mathematical details of how the augmented model is derived.

ii) [8%] Now use the augmented model of task (i) to write down the optimal control problem formulation (the objective function and the constraints) for designing a linear MPC having the integral action feature.

iii) Now, in order to make the MPC problem with integral action, you should convert the optimal control problem of task (ii) into a standard QP problem given by equation 4 in an efficient way (say using kronecker product formulation).

$$\min_{z} \quad \frac{1}{2}z^T H z + c^T z,$$
$$s.t \qquad \qquad \qquad \qquad \qquad (4)$$
$$A_e z = b_e,$$
$$z_L \leq z \leq z_H.$$

The vector of unknowns, z, must include the following elements from the **augmented model** in their respective places: [inputs, states, error and measurement].

Answer the following for task (iii):

(a) [10%] Find the expression for matrix H and vector c.

(b) [12%] Find the expression for matrix A_e and vector b_e.

Problem 2

A finite horizon MPC cannot guarantee stability even if a feasible solution exists. To guarantee closed loop stability MPC should be formulated as having an infinitely long prediction horizon, which obviously possess some practical difficulties. Assuming that your optimal control with finite horizon length N is given by,

$$\min_{(u_0, u_1 \dots \dots \dots \dots u_{N-1})} \quad J = \sum_{k=0}^{N-1} q(x_k, u_k)$$

$$\begin{aligned} s.t. \quad & x_{k+1} = f(x_k, u_k) \;, \quad k = 0,1,2, \dots \dots \dots, N-1 \;\rightarrow \textit{nonlinear process model}, \qquad x_0 \textit{ known} \\ & y_k = g(x_k, u_k) \;, \quad k = 0,1,2, \dots \dots \dots, N-1 \;\rightarrow \textit{measurement equations} \\ & h_i(x) - b_i = 0 \;, \quad i = 1,2, \dots \dots \dots, m \;\rightarrow \textit{nonlinear equality constraints} \\ & I_j(x) - c_j \leq 0 \;, \quad j = 1,2, \dots \dots \dots, r \;\rightarrow \textit{nonlinear inequality constraints} \end{aligned} \qquad (6)$$

Tasks

[8%] Show with mathematical descriptions as to how terminal costs and terminal constraints can be utilized for improving the stability of a finite horizon MPC.

Problem 3

Tasks

(i) [7%] What is the difference between a state feedback MPC and an output feedback MPC. Draw necessary signal flow diagrams (block diagrams) to illustrate your reasoning.

(ii) [7%] Explain the weighted sum method for solving a multi objective optimization problem with necessary diagram. Also write down its disadvantages if there are any.

Roshan Sharma
18th November 2022, Porsgrunn, Norway